Series &RQPS

Set – 5

प्रश्न-पत्र कोड
Q.P. Code **65(B)**

अनुक्रमांक
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- *कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **23** हैं ।*
- *कृपया जाँच कर लें कि इस प्रश्न-पत्र में **38** प्रश्न हैं ।*
- *प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।*
- ***कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।***
- *इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।*
- *Please check that this question paper contains **23** printed pages.*
- *Please check that this question paper contains **38** questions.*
- *Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.*
- ***Please write down the serial number of the question in the answer-book before attempting it.***
- *15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.*

गणित

(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

MATHEMATICS

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे | *अधिकतम अंक : 80*

Time allowed : 3 hours | *Maximum Marks : 80*

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General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **38** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*

(iii) *In **Section A,** Questions no. **1** to **18** are multiple choice questions (MCQs) and questions number **19** and **20** are Assertion-Reason based questions of **1** mark each.*

(iv) *In **Section B,** Questions no. **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.*

(v) *In **Section C,** Questions no. **26** to **31** are short answer (SA) type questions, carrying **3** marks each.*

(vi) *In **Section D,** Questions no. **32** to **35** are long answer (LA) type questions carrying **5** marks each.*

(vii) *In **Section E,** Questions no. **36** to **38** are case study based questions carrying **4** marks each.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 3 questions in Section C, 2 questions in Section D and 2 questions in Section E.*

(ix) *Use of calculators is **not** allowed.*

SECTION A

This section comprises multiple choice questions (MCQs) of 1 mark each.

1. If $A = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$, then the value of x, for which A is an identity matrix, is

(A) $\frac{\pi}{2}$ (B) π

(C) 0 (D) $\frac{3\pi}{2}$
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2. If the matrix $A = \begin{bmatrix} 0 & 5 & -7 \\ a & 0 & 3 \\ b & -3 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then the values of 'a' and 'b' are :

(A) $a = 5, b = 3$ (B) $a = 5, b = -7$

(C) $a = -5, b = -7$ (D) $a = -5, b = 7$

3. If $\begin{vmatrix} x+2 & x-4 \\ x-2 & x+3 \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 1 & 3 \end{vmatrix}$, then the value of x is :

(A) 1 (B) 2

(C) -2 (D) -1

4. If $\begin{bmatrix} 8 & 14 \\ 9 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} X$, then matrix X is :

(A) $\begin{bmatrix} 3 & 7 \\ 2 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 0 \\ 7 & 3 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 0 \\ -3 & 7 \end{bmatrix}$

5. The value of k, for which $f(x) = \begin{cases} \dfrac{\sqrt{3}\cos x + \sin x}{3x + \dfrac{\pi}{2}}, & x \neq -\dfrac{\pi}{3} \\ k, & x = -\dfrac{\pi}{3} \end{cases}$

is continuous at $x = -\dfrac{\pi}{3}$, is :

(A) $\dfrac{2}{3}$ (B) $-\dfrac{2}{3}$

(C) $\dfrac{3}{2}$ (D) 6

6. Let the vectors $\vec{a}$ and $\vec{b}$ be such that $|\vec{a}| = \sqrt{3}$ and $|\vec{b}| = \frac{2}{\sqrt{3}}$, then $\vec{a} \times \vec{b}$ is a unit vector, if the angle between $\vec{a}$ and $\vec{b}$ is :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

7. If $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$, then the projection of $(\vec{c} - \vec{b})$ along $\vec{a}$ is :

(A) 15 (B) 5

(C) $\frac{2}{3}$ (D) 1

8. The angle between the lines

$\frac{x+1}{2} = \frac{2-y}{-5} = \frac{z}{4}$ and $\frac{x-3}{1} = \frac{y-7}{2} = \frac{5-z}{3}$ is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

9. The Cartesian equations of a line are given as

$$6x - 2 = 3y + 1 = 2z - 2$$

The direction ratios of the line are :

(A) 2, – 1, 3 (B) 1, – 2, – 3

(C) 1, 2, 3 (D) 3, 1, 2

10. The solution set of the inequation $2x + 3y < 6$ is :

(A) open half-plane not containing origin

(B) whole xy-plane except the points lying on the line $2x + 3y = 6$

(C) open half-plane containing origin

(D) half-plane containing the origin and the points lying on the line $2x + 3y = 6$

11. The maximum value of the objective function $z = 3x + 5y$ subject to the constraints $x \geq 0, y \geq 0$ and $4x + 3y \leq 12$ is :

(A) 15 (B) 29

(C) 9 (D) 20

12. If the points A(3, – 2), B(k, 2) and C(8, 8) are collinear, then the value of k is :

(A) 2 (B) – 3

(C) 5 (D) – 4

13. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then $(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$ is equal to :

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$

(C) $-\frac{1}{2}$ (D) $-\frac{3}{2}$

14. $\int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx$ is equal to :

(A) $\cot x + \tan x + c$ (B) $-\cot x + \tan x + c$

(C) $\cot x - \tan x + c$ (D) $-\cot x - \tan x + c$

15. The solution of the differential equation $\frac{dy}{dx} = 1 - x + y - xy$ is :

(A) $\log |1 + y| = x - \frac{x^2}{2} + c$ (B) $\log |1 + y| = -x + \frac{x^2}{2} + c$

(C) $e^y = x - \frac{x^2}{2} + c$ (D) $e^{(1+y)} = -x + \frac{x^2}{2} + c$

16. The degree of the differential equation

$$x\left(\frac{d^2y}{dx^2}\right)^3 + y\left(\frac{dy}{dx}\right)^4 + y^5 = 0 \text{ is :}$$

(A) 2 (B) 3

(C) 4 (D) 5

17. The integrating factor of the differential equation $\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x$ is :

(A) $e^{\sec x}$ (B) $\sec x + \tan x$

(C) $\sec x$ (D) $\cos x$

18. The probabilities of A, B and C solving a problem are $\frac{1}{3}, \frac{1}{5}$ and $\frac{1}{6}$ respectively. The probability that the problem is solved, is :

(A) $\frac{4}{9}$ (B) $\frac{5}{9}$

(C) $\frac{1}{90}$ (D) $\frac{1}{3}$

*Questions number **19** and **20** are Assertion and Reason based questions. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. *Assertion (A) :* $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{6}$

Reason (R) : $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$

20. *Assertion (A) :* If the side of a square is increasing at the rate of 0·2 cm/s, then the rate of increase of its perimeter is 0·8 cm/s.

Reason (R) : Perimeter of a square = 4 (side).

SECTION B

This section comprises very short answer (VSA) type questions of 2 marks each.

21. (a) Find the value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$.

OR

(b) Find the domain of the function $y = \cos^{-1}(x^2 - 4)$.

22. (a) Differentiate $\cot^{-1}(\sqrt{1 + x^2} + x)$ w.r.t. x.

OR

(b) If $(\cos x)^y = (\cos y)^x$, find $\frac{dy}{dx}$.

23. Find the intervals on which the function $f(x) = 10 - 6x - 2x^2$ is (a) strictly increasing (b) strictly decreasing.

24. Show that of all rectangles inscribed in a given circle, the square has the maximum area.

25. Find :

$$\int \operatorname{cosec}^3 (3x+1) \cot (3x+1)\, dx$$

SECTION C

This section comprises short answer (SA) type questions of 3 marks each.

26. If $x = a \cos \theta$ and $y = b \sin \theta$, then prove that $\dfrac{d^2y}{dx^2} = -\dfrac{b^4}{a^2y^3}$.

27. Find :

$$\int \frac{2x}{(x^2+1)(x^2+3)}\, dx$$

28. (a) Evaluate :

$$\int_{-6}^{6} |x+2|\, dx$$

OR

(b) Find :

$$\int \left(\frac{4-x}{x^5}\right) e^x\, dx$$

29. (a) Find the particular solution of the differential equation $2xy \frac{dy}{dx} = x^2 + 3y^2$, given that $y(1) = 0$.

OR

(b) Solve the differential equation $\frac{dy}{dx} + 2y \tan x = \sin x$, given that $y = 0$, when $x = \frac{\pi}{3}$.

30. The corner points of the feasible region determined by the system of linear constraints are A(0, 40), B(20, 40), C(60, 20) and D(60, 0). The objective function of the L.P.P. is $z = 4x + 3y$. Find the point of the feasible region at which the value of objective function is maximum and the point at which the value is minimum. Hence, find the maximum and the minimum values.

31. (a) A card is randomly drawn from a well-shuffled pack of 52 playing cards. Events A and B are defined as under :

A : Getting a card of diamond

B : Getting a queen

Determine whether the events A and B are independent or not.

OR

(b) Find the probability distribution of the number of doublets in three throws of a pair of dice.

SECTION D

This section comprises long answer (LA) type questions of 5 marks each.

32. (a) Let $A = \{x \mid x \in Z, 0 \leq x \leq 12\}$. Show that the relation $R = \{(a, b) : a, b \in A, (a - b) \text{ is divisible by } 4\}$ is an equivalence relation. Find the set of elements related to 2.

OR

(b) Let $A = \mathbb{R} - \{4\}$ and $B = \mathbb{R} - \{1\}$ and let function $f : A \to B$ be defined as $f(x) = \frac{x-3}{x-4}$ for $\forall\, x \in A$. Show that f is one-one and onto.

33. Using matrices, solve the following system of linear equations :

$$3x + 4y + 2z = 8\ ;\ 2y - 3z = 3\ ;\ x - 2y + 6z = -2$$

34. Using integration, find the area of the region bounded by the curve $y = x^2$, $x = -1$, $x = 1$ and the x-axis.

35. (a) Write the vector equations of the following lines and hence find the shortest distance between them :

$$\frac{x+1}{2} = \frac{y+1}{-6} = \frac{z+1}{1} \text{ and } \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$$

OR

(b) Find the length and the coordinates of the foot of the perpendicular drawn from the point P(5, 9, 3) to the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Also, find the coordinates of the image of the point P in the given line.

SECTION E

This section comprises 3 case study based questions of 4 marks each.

Case Study – 1

36. The relation between the height of the plant (y in cm) with respect to exposure to sunlight is governed by the relation $y = 4x - \frac{1}{2}x^2$, where x is the number of days it is exposed to sunlight.

Based on the above, answer the following questions :

(i) Find the rate of growth of the plant with respect to sunlight. *1*

(ii) What is the number of days it will take for the plant to grow to the maximum height ? *2*

(iii) What is the maximum height of the plant ? *1*

Case Study – 2

37. A cricket match is organised between two clubs P and Q for which a team from each club is chosen. Remaining players of club P and club Q are respectively sitting along the lines AB and CD, where the points are A(3, 4, 0), B(5, 3, 3), C(6, – 4, 1) and D(13, – 5, – 4).

Based on the above, answer the following questions :

(i) Write the direction ratios of vector $\overrightarrow{AB}$. *1*

(ii) Write a unit vector in the direction of $\overrightarrow{CD}$. *1*

(iii) (a) Find the angle between vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$. *2*

OR

(iii) (b) Write a vector perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{CD}$. *2*

Case Study – 3

38. A coach is training 3 players. He observes that player A can hit a target 4 times in 5 shots, player B can hit 3 times in 4 shots and player C can hit 2 times in 3 shots.

Based on the above, answer the following questions :

(i) Find the probability that all three players miss the target. *1*

(ii) Find the probability that all of them hit the target. *1*

(iii) (a) Find the probability that only one of them hits the target. *2*

OR

(iii) (b) Find the probability that exactly two of them hit the target. *2*